POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 1051 **Semester I** **Automobile Engineering** **Practicum**

Basic Automobile Engineering

A complete first-semester handbook connecting vehicle architecture, thermodynamic energy conversion, IC-engine mechanisms and the safe well-to-wheel fuel journey.

90 h
official contact hours

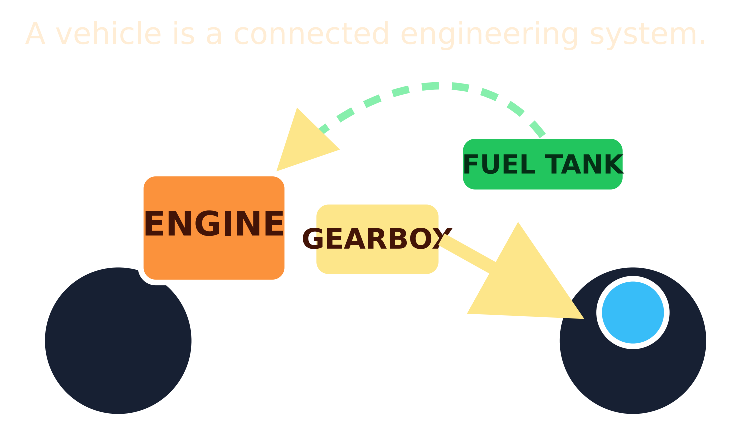
45 + 45
module-total theory + practical

4.5
credits

3:0:3:0
L:T:P:R scheme

40 + 100
CIE + ESE

24
official activities



Original technical illustration: fuel energy passes through engine and transmission to road wheels, supported by the structure.

AUTHORITY AND LIMITS

Official Source Declaration and Verified Inconsistencies

This handbook uses only the supplied official SITTTR Kerala **Diploma Curriculum Revision 2026** syllabus PDF for **1051 Basic Automobile Engineering** as curriculum authority. No Revision 2021 subject content was merged. Supporting explanations, current plant-location examples, original diagrams and practice questions are not additional official syllabus.

Assessment presentation: Page 1 states CIE 40 and ESE 100. Page 2 splits this into Theory 20+60 and Practical 20+40, producing the same totals. Both official presentations are retained.

Ambiguous page-2 value: The value 6 appears below the heading “Self-learning (SS)/Week”, while the stated teaching scheme is 3:0:3:0 and module totals are 45 L + 45 P. The PDF does not explain whether 6 is intended as self-learning or total weekly engagement. It is preserved as an ambiguity and is not added to the 90 contact hours.

HMV wording error: The detailed outcome for “Measurement of Vehicle Geometric Dimensions-HMV” says “Light Motor Vehicle (HMV)”. This handbook uses Heavy Motor Vehicle, consistent with the abbreviation and major-topic table.

Third Law typo: “Thrd Law” is corrected only in spelling to “Third Law”.

Module IV detail omission: The major-topic table lists six practicals, but the supplied detailed table shows learning-outcome rows only for the first two before the PDF ends. All six are retained from the official major-topic table.

Not specified in the supplied official syllabus PDF: official textbooks, references, websites, detailed equipment list, exact standard test methods, manufacturer service values and an official model question paper.

COURSE DATA

Course Dashboard

4

Course Outcomes

4

modules

140

total marks

Understand

overall CO level

Official teaching and assessment scheme								
L	T	P	S/R	Printed SS value	Total h	Credits	Theory CIE/ESE	Practical CIE/ESE
3	0	3	0	6 — ambiguous	90	4.5	20 / 60	20 / 40

Course description

The course introduces automobile evolution and architecture, thermodynamic energy conversion, IC-engine construction and operation, and the fuel journey from crude oil to the vehicle tank. Theory is tied to identification, measurement and safe workshop observation.

Prerequisite

Basic knowledge in science — Secondary Education.

> Define the system boundary, isolate the vehicle/apparatus and record the condition before any observation.

Official CO–PO mapping

Course articulation matrix											
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	–	–	–	–	2	–	–	–	–	–
CO2	3	2	–	–	–	–	–	–	–	–	–
CO3	3	2	–	2	–	–	–	–	2	–	–
CO4	3	–	–	–	3	3	3	2	–	–	–
Total	12	4	–	2	3	5	3	2	2	–	–
Correlation level	3	2	–	2	3	2.5	3	2	2	–	–

LEARNING METHOD

How to Use This Handbook

1 · Understand

Read the concept, definition and assumptions; use Malayalam support to confirm meaning.

2 · Observe

Study the diagram and trace power, heat, mass or force direction.

3 · Demonstrate

Follow approved procedure, record actual data, calculate with units and state a defensible result.

4 · Check

Attempt expected questions before opening model answers.

5 · Record

Use experiment cards as record format and replace blanks with your own laboratory data.

6 · Revise

Use the Bank, visual library and quick-revision checklist before examinations.

OFFICIAL WORDING

Objectives and Course Outcomes

Course Objectives

1. To describe the evolution, industry players, and physical architecture of automobiles.
2. To explain fundamental thermodynamic laws and cycles related to energy conversion.
3. To identify internal combustion engine components and their operating principles.

4. To outline fuel properties, safety standards, and storage requirements.

Course Outcomes

CO1 — Understand: Describe the historical development of automobiles, major Indian manufacturers, and the fundamental layouts of vehicle chassis and dimensions.

CO2 — Understand: Explain the basic principles of thermodynamics, including system types, laws of energy conservation, and theoretical engine cycles.

CO3 — Understand: Differentiate between internal engine components and the operating principles of 2-stroke, 4-stroke, Petrol, and Diesel engines.

CO4 — Understand: Discuss the journey of fuel from crude oil to the vehicle tank, including fuel grades, safety protocols, and BS-VI standards.

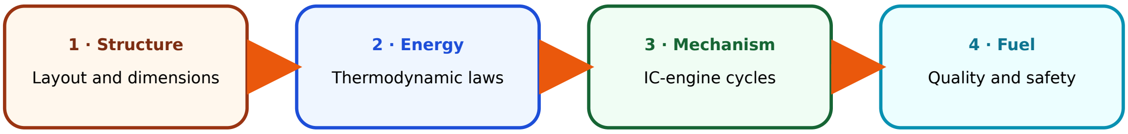
Malayalam support: Component □□□□□□ □□□□□□□□□□□□□□□□□□□ □□□□□□ □□□□. Vehicle layout, energy conversion, engine mechanism, fuel safety □□□□□□ □□□□□□□□□□□ □□□□□ workshop observation-□□□□□□ explain □□□□□□□.

COVERAGE MAP

Curriculum Coverage Map and Learning Roadmap

Official module coverage

Module	Official topics	L + P	CO	Taxonomy	Handbook
I · Introduction to Automobile	Evolution, manufacturers and plants/products; chassis/body; four-wheeler layouts; dimensions, weight and CG; engine mounts; six activities.	10 + 10	CO1	Remember / Understand	Module I
II · Basics of Thermodynamics	System and properties; equilibrium; four laws; processes; Carnot/Otto/Diesel P–V cycles; six activities.	12 + 12	CO2	Remember / Understand	Module II
III · Engine Basics	Components and terminology; four-stroke petrol/diesel; two-stroke, scavenging and port timing; SI/CI comparison; six activities.	12 + 12	CO3	Remember / Understand	Module III
IV · Fuels	Origin/refining; storage/handling; petrol and diesel properties; flash/fire point; BS-VI, biofuel and filling-station layout; six activities.	11 + 11	CO4	Remember / Understand	Module IV



Learn what carries the vehicle, then how energy behaves, how the engine converts it, and how fuel reaches and operates the system safely.

MODULE I

Introduction to Automobile

Vehicle engineering begins by identifying the structure, aggregate locations, dimensions and interfaces that make a complete road vehicle.

10 L + 10 P **CO1** **Remember / Understand**

Learning objectives

- Trace automobile evolution.
- Recognise major Indian manufacturers and product categories.
- Classify chassis/body and drivetrain layouts.
- Measure principal dimensions and inspect mounts.

Core question

Where are the major aggregates, how does torque reach the road, and what structure carries the resulting loads?

Workshop habit

Immobilise the vehicle, define the inspection boundary and record observed—not assumed—construction.

1.1 Evolution of automobiles — abroad and in India—

Road transport evolved through human and animal power, steam vehicles, early electric vehicles and internal-combustion vehicles. The late nineteenth and twentieth centuries brought reliable petrol and diesel engines, pneumatic tyres, mass production, all-steel bodies, hydraulic brakes, automatic transmissions and electronic control.

Modern development adds fuel injection, emission after-treatment, active safety, hybrid powertrains, battery-electric drives, connectivity and software-defined functions.

In India, early imported vehicles were followed by assembly and licensed manufacture, then indigenous commercial and passenger-vehicle production, supplier-network growth, small-car mass motorisation, emission-stage upgrades and rapid expansion of two-wheelers, three-wheelers, utility vehicles, buses, trucks and electrified products.

> **Malayalam support:** Automobile evolution □□□□□□ engine □□□□□□ □□□□□□ □□□□. Manufacturing method, body construction, brakes, tyre, electronics, emission control, safety, fuel/energy source □□□□□□□□□□ □□□□□□□□ □□□□□□□□□.

> **Industrial view:** Every technology step creates new drawing, wiring, test, service and safety requirements. A designer must understand the vehicle as a system, not as isolated parts.

1.2 Major Indian manufacturers, plants and core products−

The syllabus names Hero, Bajaj, Maruti Suzuki, Tata Motors, Ashok Leyland and Mahindra. The following plant list is a **supplementary July 2026 learning snapshot** from official manufacturer material; plants and product allocations can change, so current official information takes priority.

Manufacturer overview — supplementary, not official syllabus wording

Manufacturer	Vehicle focus	Representative Indian manufacturing locations	Core products
Hero MotoCorp	2W	Dharuhera, Gurugram, Haridwar, Neemrana, Halol, Chittoor	Motorcycles and scooters
Bajaj Auto	2W / 3W	Waluj, Chakan, Akurdi, Pantnagar	Motorcycles, scooters, passenger/goods three-wheelers and quadricycle
Maruti Suzuki	LMV	Gurugram, Manesar, Kharkhoda; Hansalpur in Gujarat	Passenger cars and utility vehicles
Tata Motors	LMV / HMV	Jamshedpur, Pune, Lucknow, Pantnagar, Sanand, Dharwad	Passenger vehicles, trucks, buses and commercial vehicles
Ashok Leyland	HMV	Ennore, Hosur, Alwar, Bhandara, Pantnagar and other facilities	Trucks, buses, engines and defence/special vehicles
Mahindra & Mahindra	3W / LMV / HMV	Chakan, Nashik, Kandivali, Zaheerabad, Haridwar and other plants	Utility vehicles, SUVs, commercial vehicles and selected three-wheelers

Vehicle-category abbreviations: 2W — two-wheeler; 3W — three-wheeler; LMV — light motor vehicle; HMV — heavy motor vehicle.

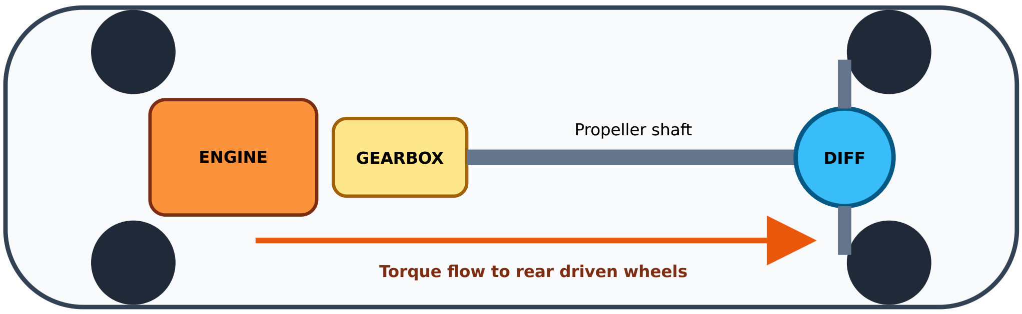
Malayalam support: Plant location □□□□□□□□□□ company name □□□□□□ □□□□□□□□□. State/location, □□□□□□ vehicle category, core product □□□□□□ table □□□□□□□□□ □□□□□□□□□. Product allocation □□□□□; official company source verify □□□□□□□□.

1.3 Chassis, body and four-wheeler layout−

Chassis is the load-carrying arrangement that supports and locates engine and powertrain, suspension, steering, brakes, fuel or energy storage and body. **Body** forms the occupant or cargo enclosure and external shape. In body-on-frame construction a separate frame carries major loads; in monocoque or unibody construction the welded body shell is the main structure. Many modern vehicles also use front and rear subframes.

Common four-wheeler layouts

Layout	Power path	Advantages	Limitations / use
Front-engine FWD	Engine → transaxle → front wheels	Compact, no long propeller shaft, good passenger space	High front-axle load; common LMV
Front-engine RWD	Engine → gearbox → propeller shaft → rear differential	Handles higher torque and load well; balanced duties	Driveline tunnel and more components; cars and HMVs
Rear-engine RWD	Rear engine/transmission → rear wheels	Short power path, useful packaging in buses or special cars	Cooling, service and packaging challenges
AWD / 4WD	Power distributed to both axles	Higher traction	Mass, cost, losses and control complexity



Power-flow diagram for a typical front-engine RWD vehicle.

Malayalam support: Engine □□□□□□□ □□□□□□□□□□□□□□□ □□□□□□ FWD □□□□□ □□□□□□□□□□□□□□□□. Gearbox □□□□□□□ propeller shaft rear differential-□□□□□□ □□□□□□□□□□□□□□□□□□ □□□ RWD layout □□□.

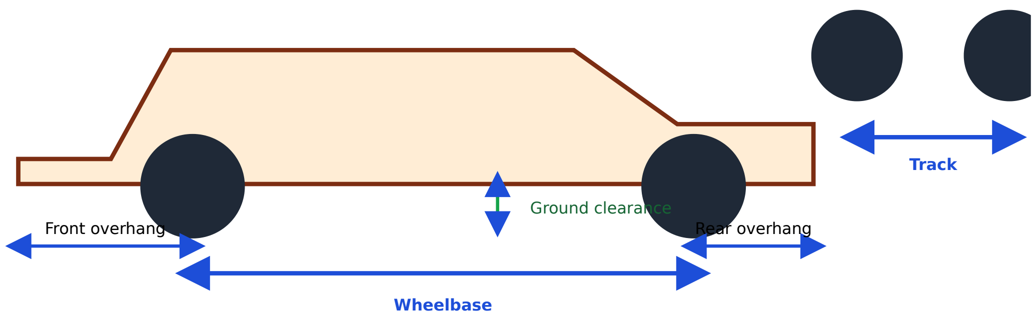
1.4 Vehicle dimensions, weights and centre of gravity−

Geometric terms

- **Wheelbase:** longitudinal distance between front and rear axle centre lines.
- **Track:** lateral distance between wheel centre planes on the same axle.
- **Front/rear overhang:** axle centre line to the corresponding vehicle extremity.
- **Ground clearance:** minimum ground-to-lowest specified point distance under stated condition.

Mass and stability terms

- **Kerb weight:** ready-to-operate vehicle mass under the applicable specified condition, without payload.
- **GVW:** total vehicle weight or mass at the specified gross condition.
- **CG height:** vertical location of the vehicle centre of gravity; a higher CG increases load transfer and rollover tendency.



Measure between axle or wheel centre references, not convenient tyre edges.

Consistency check

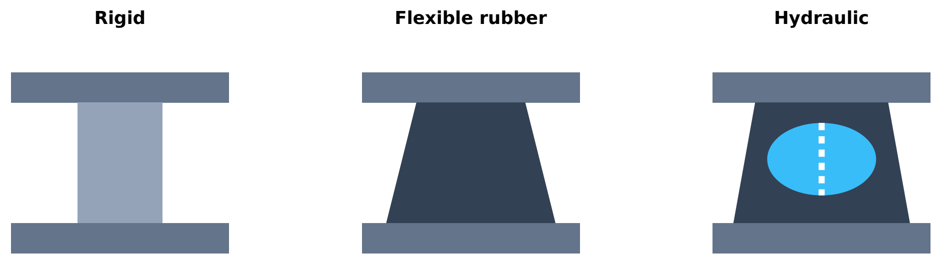
For a vehicle with front overhang 850 mm, wheelbase 2600 mm and rear overhang 750 mm, estimated overall length = 850 + 2600 + 750 = **4200 mm**. A large mismatch indicates wrong reference points or measurement error.

Common error: Track is not the external tyre-to-tyre width. State the centre-plane convention.

1.5 Engine mounts — purpose, types and selection−

Engine mounts carry static weight, resist acceleration, braking and torque reaction, maintain driveline alignment, limit movement and reduce vibration transmitted to the structure. Selection balances stiffness, damping, displacement, durability, temperature, oil exposure and fail-safe restraint.

Mount comparison			
Type	**Construction or behaviour**	**Typical use**	**Failure clues**
Rigid	Metal-to-metal or nearly rigid location; little isolation	Special fixtures and stationary applications	High transmitted vibration; looseness or crack
Flexible / rubber	Elastomer carries load and allows controlled movement	Common automotive mounting	Cracks, separation, sag, oil attack
Hydraulic	Elastomer plus fluid chamber or orifice for frequency-dependent damping	Passenger vehicles needing improved isolation	Fluid leak, collapse, excessive movement



Simplified sections; actual mount geometry and fluid paths are manufacturer-specific.

Malayalam support: Soft mount vibration □□□□□□□□□□, □□□□□□ engine movement □□□□□□. Hard mount movement □□□□□□□□□□□□□□□, vibration body-□□□□□□□ □□□□□□□ transmit □□□□□□□□. Selection □□□ compromise □□□.

Module I summary

Structure carries loads and fixes aggregate relationships.

Layout classification follows actual power flow.

Dimensions require defined reference points and level conditions.

Expected / Practice Questions

Very Short **Define chassis.**

The chassis is the structural system that supports and locates the vehicle aggregates and transmits service loads.

Short Answer **Differentiate chassis and body.**

The chassis is the load-carrying foundation for powertrain, suspension, steering and brakes; the body forms the enclosure or cargo shape and may be separate or structurally integrated in a monocoque.

Comparison **Compare FWD and RWD layouts.**

FWD groups engine, transaxle and driven wheels at the front, saving space and the propeller shaft. RWD sends torque to the rear axle, often suiting higher torque and load but requiring a propeller shaft and tunnel.

Descriptive **Explain wheelbase, track, overhang and ground clearance.**

Wheelbase is axle-centre distance longitudinally; track is distance between wheel centre planes on an axle; overhang is axle centre to vehicle extremity; ground clearance is minimum ground-to-low-point distance under a stated loading condition.

Application **Why are engine mounts not selected only by engine mass?**

They must also control torque reaction, vibration frequencies, displacement limits, heat or oil exposure, durability and driveline alignment.

Diagram **Draw a front-engine rear-wheel-drive power flow.**

Show engine → clutch or torque converter → gearbox → propeller shaft → differential or final drive → axle shafts → rear wheels.

MODULE II

Basics of Thermodynamics

Thermodynamics provides the language for tracking mass, heat, work and state changes in engines, cooling systems and fuel-energy conversion.

12 L + 12 P | CO2 | Remember / Understand

Learning objectives

- Define system, surroundings and boundary.
- Classify systems and properties.
- Explain all four thermodynamic laws.
- Read basic P–V cycles.

Sign convention

This handbook uses heat into a closed system positive and work done by the system positive: **Q − W = ΔU**.

Absolute temperature

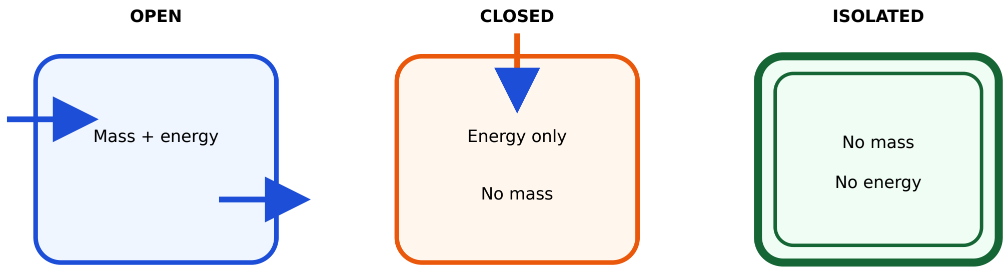
Use kelvin for thermodynamic efficiency ratios: T(K) = t(°C) + 273.15.

2.1 System, surroundings, boundary and universe−

The **system** is the quantity of matter or region chosen for study. The **surroundings** are everything outside it. The **boundary** separates them and may move or be imaginary. System plus surroundings form the **universe** for the analysis.

System classification with automotive examples

Type	Mass crossing?	Energy crossing?	Example with stated boundary
Open / control volume	Yes	Yes	Running engine, radiator or turbocharger
Closed / control mass	No	Heat or work may cross	Gas trapped in cylinder while both valves are closed
Isolated, ideal	No	No	Ideal insulated sealed model; real systems only approximate



Classification depends on the boundary and the period being analysed.

Malayalam support: Component-□□ □□□□□□□□□ □□□ classification □□□□. Boundary □□□□□ □□□□□□□□□□□□□, □□□ □□□□□□□□ analyse □□□□□□□□□□ □□□□□□□□□□□□□□ open □□□□□□□□□□ closed classification □□□□□.

2.2 Properties, state and equilibrium—

A property is a measurable characteristic. **Intensive** properties are independent of system size; **extensive** properties scale with mass or quantity. Specific properties such as specific volume are extensive properties divided by mass and are therefore intensive.

Property classification

Intensive	Extensive	Automotive note
Temperature, pressure, density, viscosity	Mass, volume, total internal energy, total enthalpy	Engine-oil viscosity must be stated at a temperature.

The **state** is specified by sufficient independent properties. **Thermal equilibrium** means no temperature-driven heat transfer; **mechanical equilibrium** means no unbalanced pressure or force tendency; **chemical equilibrium** means no net reaction or composition tendency. Thermodynamic equilibrium requires all relevant forms.

Malayalam support: 1 litre oil-□□ 2 litre □□□□□□□□□□ mass, volume □□□□□—extensive. □□□ condition-□ density, temperature □□□□□□□□□□□□—intensive.

2.3 Zeroth, First, Second and Third Laws—

Zeroth Law

Establishes the transitive nature of thermal equilibrium and the basis of thermometers.

First Law

Energy is conserved. For a closed system: $Q - W = \Delta U$. Enthalpy is **H = U + pV**.

Second Law

Heat has direction; real processes create entropy. A cyclic heat engine must reject heat and cannot be 100% efficient.

Third Law

Entropy of a perfect crystal approaches a minimum as temperature approaches 0 K; absolute zero cannot be reached by a finite practical process.

Worked energy balance

Problem: A closed gas receives 900 J heat and does 350 J work. Find ΔU.

Formula: $Q - W = \Delta U$. **Substitution:** $900 - 350 = 550$ J. **Answer:** Internal energy increases by **550 J**.

Heat-engine efficiency

If an engine receives $Q_H = 1000$ kJ and rejects $Q_L = 650$ kJ, $W = 350$ kJ and $\eta = W/Q_H = 0.35 =$ **35%**. Energy balance is satisfied: $1000 = 350 + 650$ kJ.

Common error: “Efficiency below 100% only because of friction” is incomplete. Even an ideal reversible engine between finite temperatures has $\eta < 100\%$.

Malayalam support: First Law energy conserve □□□□□□□□□□□. Second Law energy conversion-□□□□ direction, quality limit □□□□□□ □□□□□□□□□□□□□□□□. Radiator/exhaust heat rejection real engine-□ □□□□□□□□□□□□□□□□□.

2.4 Basic thermodynamic processes—

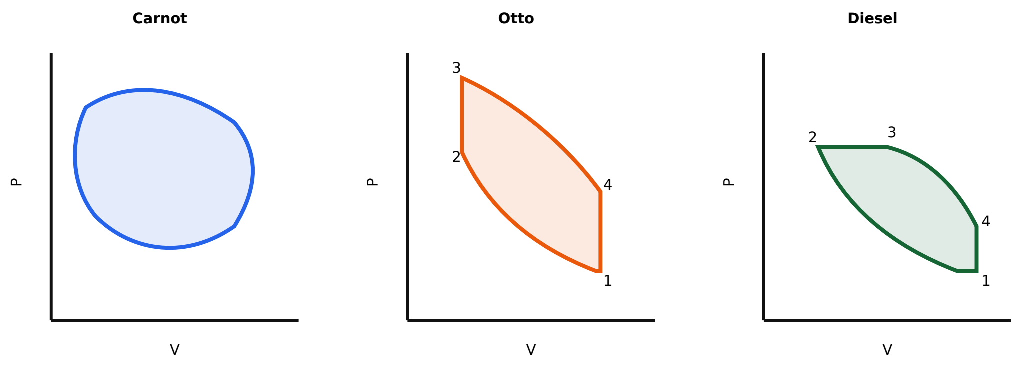
Basic processes

Process	Fixed quantity	Typical P–V appearance	Basic example
Isochoric	Volume constant	Vertical line	Idealised heat addition in Otto cycle
Isobaric	Pressure constant	Horizontal line	Idealised heat addition in Diesel cycle
Isothermal	Temperature constant	Hyperbolic for ideal gas	Slow expansion with heat exchange
Adiabatic	$Q = 0$ idealisation	Steeper curve than isothermal	Rapid compression or expansion approximation

These names describe a constraint, not automatically a complete engine stroke. Real engine processes have heat transfer, pressure loss and finite combustion duration.

Malayalam support: Isochoric = volume constant; Isobaric = pressure constant; Isothermal = temperature constant; Adiabatic = heat transfer □□□□□□□□□ idealise □□□□□□□□□□. Real engine process exact ideal process □□□□.

2.5 Carnot, Otto and Diesel P–V cycles−



Idealised diagrams: Otto heat addition is constant-volume; Diesel heat addition is constant-pressure; Carnot uses reversible isothermal and adiabatic processes.

Carnot efficiency: $\eta_{max} = 1 - T_L/T_H$, with absolute temperatures in kelvin. It is a theoretical upper bound, not the efficiency formula for a real vehicle engine.

Carnot limit example

For TH = 800 K and TL = 300 K: ηmax = 1 − 300/800 = 0.625 = **62.5%**. A real engine operating between comparable limits will be lower.

Module II summary

Define the boundary before classifying a system.

Track heat, work and stored energy with a stated sign convention.

Ideal cycles explain trends; real engines include irreversibility and losses.

Expected / Practice Questions

Very Short **Define a thermodynamic boundary.**

A real or imaginary surface separating the selected system from its surroundings.

Comparison **Differentiate intensive and extensive properties.**

Intensive properties do not depend on quantity, such as temperature or density. Extensive properties scale with quantity, such as mass, volume and total energy.

Short Answer **State the Zeroth Law in practical words.**

Bodies separately in thermal equilibrium with the same reference body have equal temperature, enabling temperature measurement.

Short Answer **State the First Law for a closed system.**

With work done by the system positive: $Q - W = \Delta U$.

Descriptive **Why is 100% heat-engine efficiency impossible?**

The Second Law requires some heat rejection to a lower-temperature sink in a cyclic heat engine; therefore QL cannot be zero in a real cycle.

Diagram **Sketch and identify Otto and Diesel P–V cycles.**

Otto: two isentropic and two constant-volume processes. Diesel: two isentropic, one constant-pressure heat-addition and one constant-volume heat-rejection process.

MODULE III

Engine Basics

An internal-combustion engine converts fuel energy into gas pressure and then into crankshaft torque through the piston–connecting-rod–crank mechanism.

12 L + 12 P | CO3 | Remember / Understand

Learning objectives

- Identify engine components.
- Use correct geometry terms.
- Explain petrol and diesel four-stroke cycles.
- Explain two-stroke scavenging and SI–CI differences.

Force path

Combustion pressure → piston → gudgeon pin → connecting rod → crankpin → crankshaft torque.

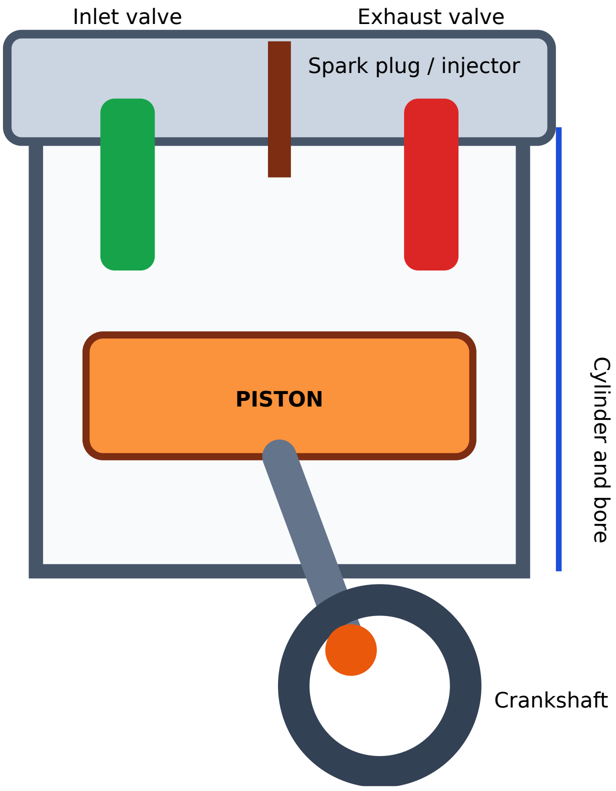
Gas path

Intake system → inlet valve or port → cylinder → exhaust valve or port → exhaust system.

3.1 Major engine components and functions−

Core components

Component	Function	Motion or load
Cylinder block	Houses cylinders, coolant and oil passages and supports the crankshaft or main structure	Stationary; pressure, thermal and bearing loads
Cylinder head	Closes cylinder, forms chamber and carries valves, ports, spark plug or injector	Stationary; thermal and pressure load
Piston and rings	Receives gas force, seals chamber, controls oil and transfers heat	Reciprocating; side thrust and thermal load
Connecting rod	Transfers force between piston pin and crankpin	Oscillating; alternating tension and compression
Crankshaft	Converts reciprocating action to rotation and delivers torque	Rotating; bending and torsion
Valves	Control intake and exhaust flow in four-stroke engines	Reciprocating; spring, impact and thermal load



Part labels and functional force path for a basic single-cylinder engine.

Malayalam support: Piston linear motion □□□□□□□□ wheel-□□□□□□ □□□□□□□□□□□. Connecting rod, crankshaft □□□ rotary torque □□□□□□□□□. Valves gas exchange control □□□□□□□□□□.

3.2 Engine terminology and geometry−

TDC and **BDC** are piston dead-centre positions. **Stroke** is their separation. **Bore** is cylinder diameter. **Swept volume** is displaced volume; **clearance volume** remains above the piston at TDC.

Swept volume per cylinder: $V_s = (\pi/4)D^2L$

Total displacement: $V_{d,total} = nV_s$

Compression ratio: $r = (V_s + V_c)/V_c$

D and L must use consistent units. Compression ratio is dimensionless.

Worked example — engine displacement

Given: 4 cylinders, bore D = 75 mm, stroke L = 80 mm.

Required: total swept volume.

$V_s = \pi/4 \times 75^2 \times 80 = 353{,}429\ \text{mm}^3 = 353.4\ \text{cm}^3$.

Total = 4×353.4 = **$1413.7\ \text{cm}^3 \approx 1.414$ L**.

Worked example — clearance volume

For $V_s = 450\ \text{cm}^3$ and $r = 10{:}1$, $V_c = V_s/(r-1) = 450/9$ = **$50\ \text{cm}^3$**. Check: $(450+50)/50 = 10$.

Common error: Do not insert diameter into πr^2 without dividing by two. The formula $\pi D^2/4$ already accounts for radius.

Malayalam support: Bore diameter □□□; Stroke piston TDC □□□□ BDC □□□□□□□□ travel □□□. Compression ratio-□□□□ unit □□□□.

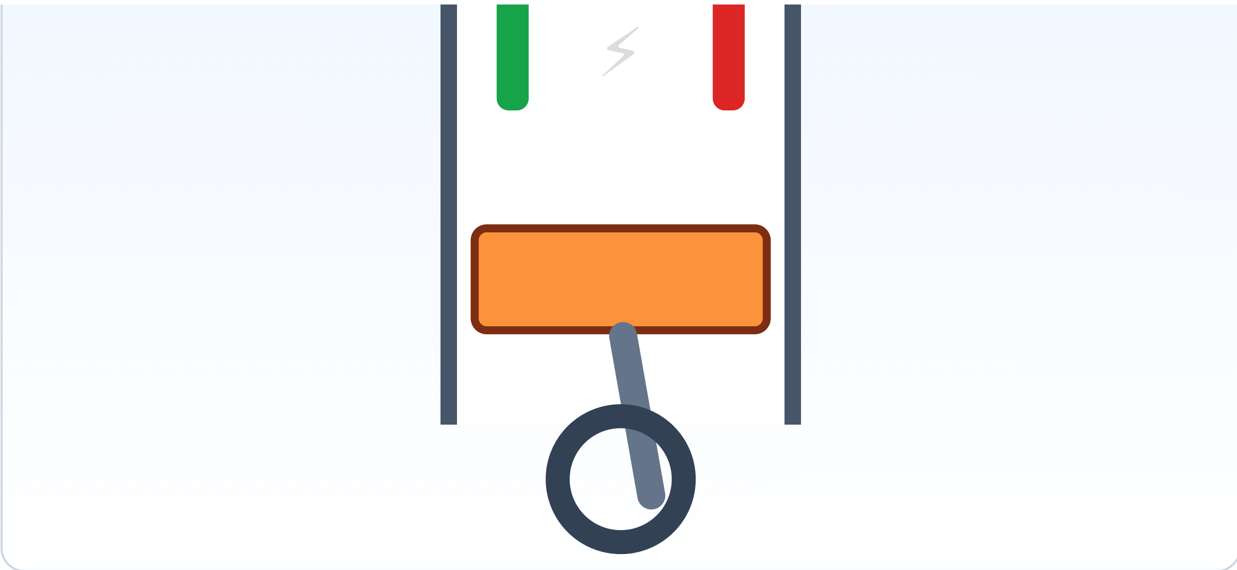
3.3 Four-stroke petrol (SI) engine—

SI four-stroke sequence				
Stroke	**Piston**	**Inlet**	**Exhaust**	**Main event**
Suction	TDC → BDC	Open	Closed	Air–fuel charge enters
Compression	BDC → TDC	Closed	Closed	Charge compressed; spark near end
Power or expansion	TDC → BDC	Closed	Closed	Combustion pressure produces work
Exhaust	BDC → TDC	Closed	Open	Products expelled

One cycle takes two crankshaft revolutions or 720°. Real valve timing opens and closes valves before or after dead centres to improve breathing; the basic sequence remains as above.

Four-stroke mechanism animation

Static four-stroke mechanism



Animation is schematic: use the stroke table for valve states and sequence. Printing shows a static mechanism.

> **Malayalam support:** Four-stroke cycle-□ crankshaft □□□□□ revolution □□□□□□□□□□□□□□□. Petrol SI engine-□ compressed charge-□□ spark plug ignite □□□□□□□□□□□.

3.4 Four-stroke diesel (CI) engine−

Diesel operation uses the same four piston strokes, but suction admits **air only**. High compression raises air temperature. Fuel is injected near the end of compression, atomises, mixes and self-ignites after a short ignition delay. Load is controlled mainly by injected fuel quantity rather than throttling a premixed charge.

Petrol SI versus diesel CI

Feature	SI petrol	CI diesel
Induction	Air–fuel charge or separately controlled direct injection depending on system	Air only during suction
Ignition	Spark plug	Self-ignition after fuel injection
Compression ratio	Generally lower	Generally higher
Fuel quality indicator	Octane or knock resistance	Cetane or ignition quality
Typical character	Higher speed, lighter systems	High low-speed torque and efficiency, heavier construction

> **High-pressure warning:** Diesel injection can penetrate skin. Never check leakage with fingers or expose anyone to spray.

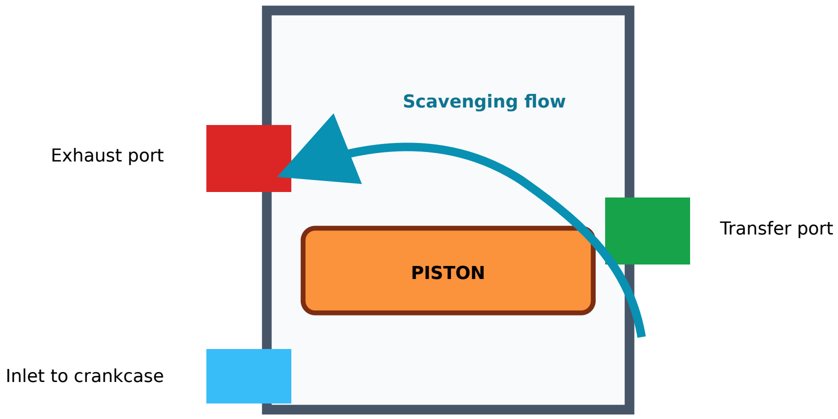
Malayalam support: Diesel engine-□ spark plug □□□□. Air □□□□□□ compress □□□□□□ temperature □□□□□□□, injector fuel spray □□□□□□□□□□□ self-ignition □□□□□□□□□□□□□.

3.5 Two-stroke petrol engine, scavenging and port timing—

A two-stroke engine completes a cycle in one crankshaft revolution. In a common crankcase-scavenged petrol engine, the rising piston compresses charge above while creating suction in the crankcase. After ignition and expansion, the descending piston compresses crankcase charge, uncovers the exhaust and then transfer ports, and fresh charge enters to displace exhaust.

Two-stroke versus four-stroke

Feature	Two-stroke	Four-stroke
Cycle	1 revolution	2 revolutions
Gas control	Ports commonly controlled by piston	Valves operated by mechanism
Power event	Every revolution ideally	Every two revolutions per cylinder
Scavenging	Critical; possible fresh-charge loss	Separate exhaust and intake strokes
Lubrication and emissions	Often more challenging	Better controlled in common designs



Fresh charge entering through the transfer port pushes exhaust toward the exhaust port; overlap can cause short-circuit loss.

Malayalam support: Scavenging □□□□□□ exhaust gas □□□□□□□□□□□ fresh charge □□□□□□□□□□□□ process □□□. Transfer, exhaust ports □□□□□□□ open □□□□ fresh charge □□□□□□□□□□□□.

Module III summary

Piston force becomes crankshaft torque through connecting rod and crank.

Geometry controls displacement and compression ratio.

SI uses spark; CI uses compression heating and injection.

Expected / Practice Questions

Very Short **Define TDC and BDC.**

TDC is the piston position nearest the cylinder head; BDC is the farthest position.

Numerical **A cylinder has bore 80 mm and stroke 90 mm. Find swept volume.**

$V_s = \pi D^2 L/4 = \pi \times (0.08)^2 \times 0.09/4 = 0.000452\ m^3 \approx 452\ cm^3$.

Numerical **For Vs = 450 cm³ and Vc = 50 cm³, find compression ratio.**

$r = (V_s+V_c)/V_c = (450+50)/50 = 10:1$.

Comparison **Compare SI and CI engines.**

SI uses spark ignition and generally lower compression ratio; CI inducts air, compresses it highly, injects diesel and self-ignites.

Descriptive **Explain the four strokes of a petrol engine.**

Suction: inlet open and piston down. Compression: both valves closed and piston up. Power: spark and combustion drive the piston down. Exhaust: exhaust valve open and piston up.

Application **Why can two-stroke engines lose fresh charge?**

During scavenging the transfer and exhaust ports may be open together, allowing part of the fresh mixture to escape with exhaust.

MODULE IV

Fuels

Fuel performance and safety depend on origin, refining, specification, clean storage, correct labelling and controlled transfer from refinery to vehicle tank.

11 L + 11 P | CO4 | Remember / Understand

Learning objectives

- Trace crude oil to petrol and diesel.
- Explain storage and vehicle filling systems.
- Distinguish fuel-quality properties.
- Apply BS-VI and fire-safety concepts.

Primary hazard

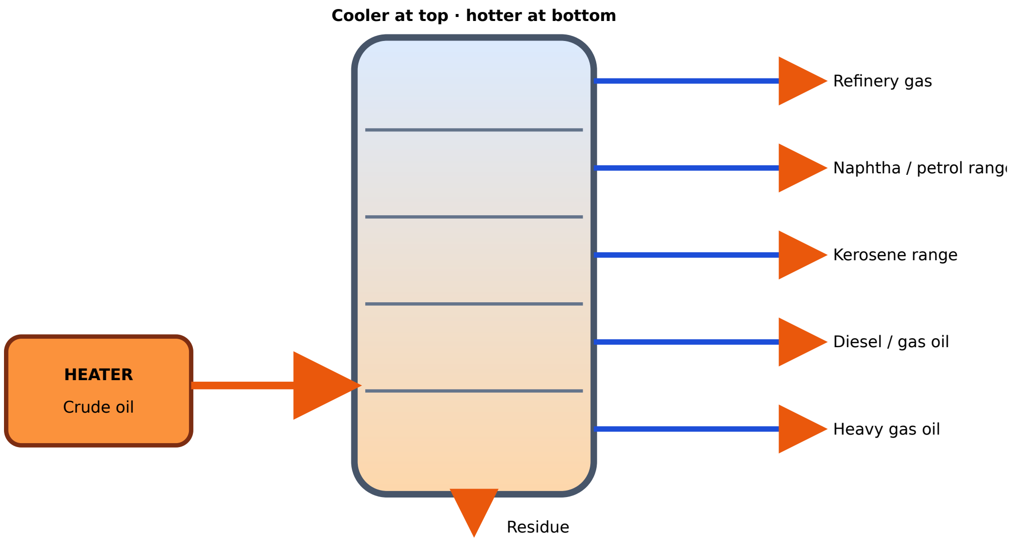
Flammable liquid and vapour require ignition control, ventilation, containment, earthing or bonding where applicable and trained emergency response.

Identification rule

Use labels, records and approved tests—not colour or smell alone.

4.1 Origin, extraction and fractional distillation−

Crude oil is a complex hydrocarbon mixture recovered from reservoirs by drilling and production systems. It is separated from water and gas, transported to a refinery, desalted and heated. In the atmospheric distillation column, a temperature gradient causes components to condense according to boiling range. Gasoline-range material and diesel or gas-oil fractions then require conversion, treating, desulphurisation, blending and testing to meet final specifications.



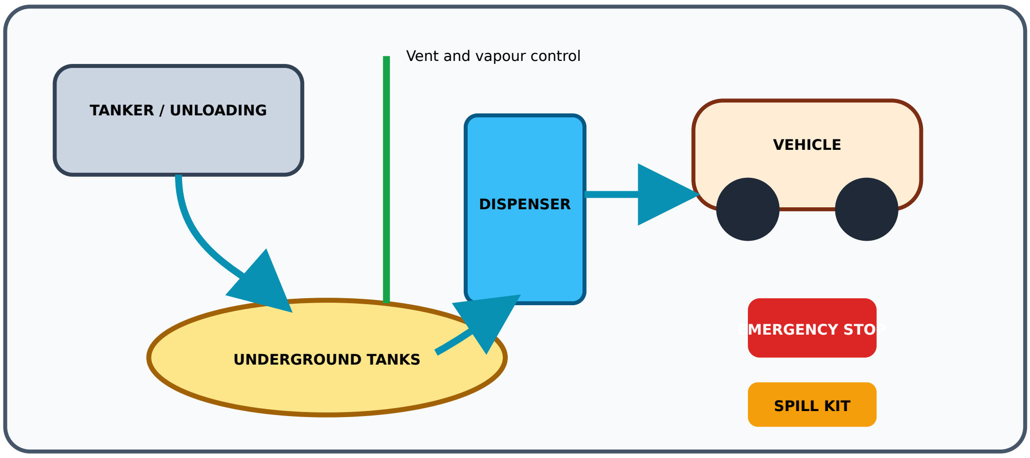
Distillation separates boiling ranges; final petrol and diesel quality also needs conversion, treating, blending and testing.

Malayalam support: Fractional distillation crude oil-□□ boiling range □□□□□□□□□□ fractions □□□ □□□□□□□□□□□□□□□□. Direct column outlet □□□□□ final petrol □□□□□□□□□□□ diesel □□□□; treating, blending, quality test □□□□□□□□□□.

4.2 Storage, handling and vehicle fuel-tank system−

Retail stations commonly receive product into underground storage tanks through controlled tanker unloading. Fill, vent or vapour, suction or delivery and monitoring systems must prevent overfill, leakage, contamination and uncontrolled vapour release. Dispensing equipment meters fuel and uses emergency shutdown and breakaway or impact controls as applicable.

The vehicle system includes cap, filler neck, anti-spill features, tank, mounting straps, vent or vapour path, pump or pick-up and supply or return lines depending on design. Petrol vapour is commonly routed to an evaporative-emission control system rather than released directly.



Conceptual retail layout; actual hazardous-area design and statutory distances require approved standards and site drawings.

Fuel handling: Stop ignition sources, control static and spillage, avoid overfill, maintain ventilation, follow the site emergency plan and never enter restricted unloading zones.

Malayalam support: Fuel tank-□□ vent □□□□□□□□□□□□; fuel level, temperature □□□□□□□□□□ pressure □□□□□□□□□□□ vacuum control □□□□□□□□. Petrol vapour direct atmosphere-□□□□□□□ □□□□□□□ EVAP system-□□□□□□□ route □□□□□□□□.

4.3 Petrol specifications — octane, volatility and calorific value−

- **Octane number:** resistance to knock in an SI engine under specified test conditions. It is not a direct measure of energy content.
- **Volatility:** tendency to vaporise. Sufficient volatility helps starting and mixing; excessive volatility can increase evaporative loss and vapour problems.
- **Calorific value:** heat released per unit mass or volume under stated test conditions. Energy available from mass m is approximately E = m × CV.

Fuel-energy example

If 0.50 kg fuel has a stated calorific value of 44 MJ/kg, chemical energy = 0.50 × 44 = **22 MJ**. This is not the mechanical output; engine efficiency and losses reduce useful work.

Malayalam support: Octane number energy content □□□□□. □□□□ petrol knock resistance □□□□. Calorific value □□□□ unit mass fuel-□ □□□□□ heat energy-□□ □□□□□□□□□□□□□□□□□□□□□.

4.4 Diesel specifications — cetane, viscosity, sulphur; flash and fire points−

- **Cetane number:** ignition-quality indicator; higher cetane generally shortens ignition delay under specified conditions.
- **Viscosity:** resistance to flow; affects injection, atomisation, leakage and lubrication of fuel-system components.
- **Sulphur content:** controlled to reduce emissions and protect after-treatment. BS-VI petrol and diesel are specified with very low sulphur, commonly maximum 10 mg/kg or ppm.
- **Flash point:** lowest specified-test temperature for momentary vapour ignition.
- **Fire point:** higher temperature at which burning continues for the specified period.

Do not confuse fuel-quality terms

Property	Main relevance	Not the same as
Octane	Petrol knock resistance	Calorific value
Cetane	Diesel ignition delay or quality	Octane
Viscosity	Flow and atomisation	Density
Flash point	Vapour ignition under test	Auto-ignition temperature

Test safety: Flash and fire-point testing must use the named standard apparatus and instructor-approved quantities. Never improvise with open cups or uncontrolled flame.

4.5 BS-VI, biofuels and ethanol blending−

BS-VI is an Indian vehicle-emission stage supported by compatible low-sulphur petrol and diesel. Vehicle control includes more precise fuelling and combustion and advanced after-treatment depending on engine type. Fuel quality and emission hardware must work together; wrong fuel or contamination can damage performance and after-treatment.

Ethanol blending adds renewable oxygenated fuel to petrol. Labels such as E10 or E20 indicate approximate ethanol percentage by volume. Compatibility depends on manufacturer design, materials, calibration and operating conditions. Use the vehicle-approved blend.

Fuel-label reading checklist

Check	Meaning	Action
Petrol or diesel	Correct engine-fuel family	Never misfuel

Check	Meaning	Action
RON or grade	Knock-resistance grade for petrol	Meet manufacturer minimum
Ethanol or biodiesel label	Blend content or category	Confirm compatibility
BS-VI or standard	Fuel-quality and emission-stage context	Use clean compliant supply

> **Malayalam support:** BS-VI, octane, ethanol blend—all different information □□□. Vehicle manual □□□□□□□□ fuel type, minimum octane, approved blend □□□□□□□ match □□□□□□□ fill □□□□□□□□□□.

Module IV summary

Distillation separates fractions; treating and blending make saleable fuels.

Fuel safety depends on vapour, ignition, spill and contamination control.

Read labels and use manufacturer-approved grade and blend.

Expected / Practice Questions

Very Short **Define octane number conceptually.**

A measure of petrol’s resistance to knock in an SI engine under specified test conditions.

Very Short **Define cetane number conceptually.**

A measure of diesel ignition quality; higher cetane generally means shorter ignition delay under test conditions.

Comparison **Differentiate flash point and fire point.**

Flash point gives momentary ignition of vapour; fire point is higher and supports sustained burning for the specified time.

Descriptive **Explain fractional distillation of crude oil.**

Heated crude enters a column with a temperature gradient; vapours condense at levels related to boiling range, separating fractions, followed by further conversion, treating and blending.

Safety **List essential fuel-station precautions.**

No smoking or ignition sources, stop engine, prevent static and overflow, follow emergency controls, maintain spill response, control tanker unloading and use trained procedures.

Application **Why is visual fuel identification unreliable?**

Colour and odour vary with dyes, additives and contamination; labels, documentation and approved testing are required.

Short Answer **What is the significance of BS-VI fuel sulphur level?**

BS-VI petrol and diesel use very low sulphur, commonly maximum 10 mg/kg or ppm, helping advanced emission-control systems operate effectively.

Application **Why must ethanol-blend compatibility be checked?**

Materials, calibration, cold-start behaviour and corrosion or water tolerance depend on vehicle design; use only the manufacturer-approved blend.

ONE-LOOK TECHNICAL BANK

Formula, Procedure and Standards Bank

Vehicle geometry

Overall length ≈ front overhang + wheelbase + rear overhang

Use one reference convention and record loading, tyre pressure and floor condition.

Engine geometry

$V_s = \pi D^2 L/4$

$V_{d,total} = nV_s$

$r = (V_s + V_c)/V_c$

Closed-system energy

$$Q - W = \Delta U$$

Heat into system positive; work done by system positive.

Heat-engine efficiency

$$\eta = W_{net}/Q_H = 1 - Q_L/Q_H$$

All energy quantities must use the same unit.

Carnot upper limit

$$\eta_{max} = 1 - T_L/T_H$$

Use absolute temperature in kelvin.

Fuel chemical energy

$$E = m \times CV$$

Mechanical output is lower because conversion efficiency is below 100%.

Universal workshop observation procedure

1. **Authorise and isolate.** Obtain instructor approval, immobilise the vehicle or apparatus and identify residual energy.
2. **Define the boundary.** State exactly which component, assembly or fluid path is being studied.
3. **Inspect before touching.** Record obvious damage, leakage, hot surfaces, pressure, sharp edges and electrical hazards.
4. **Select the method.** Use the approved drawing, tool, gauge or apparatus and verify its condition or calibration status.
5. **Observe and measure.** Use stated reference points and units; repeat where practical.
6. **Calculate and check.** Show formula, substitution, unit conversion and a physical reasonableness check.

7. **Restore and report.** Remove tools, restore guards and connections, clean the area and report abnormal findings.

> **Non-negotiable:** Never work under an unsupported vehicle, near rotating machinery, on a pressurised hot cooling system, on a live high-pressure fuel line or with unapproved open flame. Exact operating values and quantities come from the institution-approved laboratory manual or instructor.

Standards and terminology bank

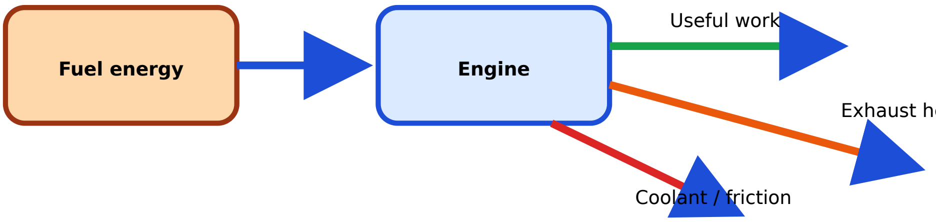
Concepts appearing in the official syllabus		
Term	**Meaning in this course**	**Engineering caution**
BS-VI	Indian Bharat Stage VI emission and compatible fuel-quality context	Do not treat it as only a fuel label; vehicle hardware, calibration and after-treatment are involved.
Octane number	Petrol knock-resistance indicator	Not calorific value.
Cetane number	Diesel ignition-quality indicator	Not a diesel version of energy content.
Flash point	Lowest test temperature for momentary vapour ignition	Method-dependent; not auto-ignition temperature.
Fire point	Test temperature for sustained burning	Requires approved apparatus and supervision.
GVW	Vehicle gross weight or mass under the applicable specified condition	Do not confuse with kerb weight.
SI / CI	Spark ignition / compression ignition	Ignition principle, not merely fuel name.

> **Malayalam support:** Formula □□□□□□□□□□□□□□□ □□□□□□ □□□□□□□. Boundary, sign convention, unit, measurement reference, safety condition □□□□□□ □□□□□□□□ □□□□□□□ calculation □□□□□□□□□□ procedure technically complete □□□.

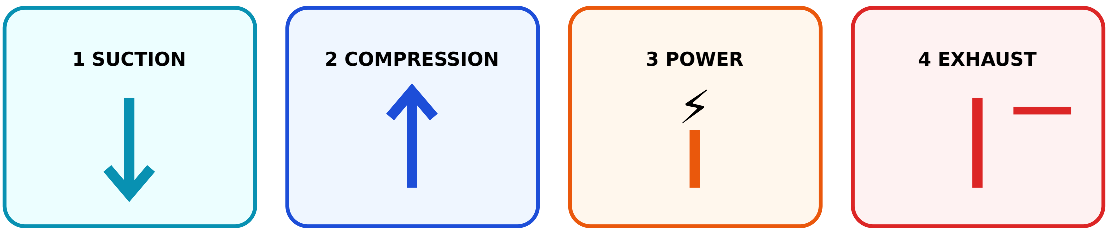
QUICK VISUAL REVISION

Diagram and Visual Library

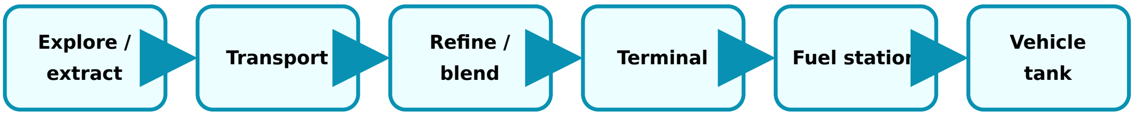
Chemical energy becomes crankshaft torque and then tractive force through the driveline.



The Second Law requires heat rejection; not all fuel energy becomes work.



One complete four-stroke cycle takes two crankshaft revolutions.



Quality documentation, clean containment and safe transfer are required at every stage.

WORKED LEARNING

Solved Examples and Demonstrations

Example 1 — vehicle length check

Given: front overhang 920 mm, wheelbase 2800 mm, rear overhang 830 mm.

Calculation: 920 + 2800 + 830 = **4550 mm**.

Interpretation: Compare with measured overall length; a major difference signals a reference or transcription error.

Example 2 — engine swept volume

Given: $D = 86$ mm, $L = 86$ mm, $n = 4$.

Per cylinder = $\pi/4 \times 86^2 \times 86 = 499{,}557\ \text{mm}^3 = 499.6\ \text{cm}^3$.

Total = $4 \times 499.6 =$ **$1998\ \text{cm}^3 \approx 2.0$ L**.

Example 3 — First Law

Given: $Q = +1.8$ kJ, $W = +0.7$ kJ.

$\Delta U = Q - W = 1.8 - 0.7 =$ **+1.1 kJ**.

The system stores more internal energy.

Example 4 — heat-engine balance

Given: QH = 2.5 MJ, W = 0.85 MJ.

QL = QH − W = 1.65 MJ; η = 0.85/2.5 = **34%**.

Example 5 — compression ratio

Given: Vs = 500 cm^3, Vc = 55.6 cm^3.

r = (500 + 55.6)/55.6 = **9.99 ≈ 10:1**.

Example 6 — fuel energy

Given: m = 0.75 kg, CV = 43 MJ/kg.

E = 0.75 × 43 = **32.25 MJ**. At an assumed 30% useful conversion only for illustration, work would be 9.675 MJ; actual efficiency must be measured, not assumed.

24 OFFICIAL ACTIVITIES

Practical / Activity / Experiment Centre

The supplied syllabus lists six practical activities in each module. The procedures below are safe educational frameworks. Institution-approved laboratory manuals, vehicle service instructions and instructor directions govern exact tools, quantities, operating conditions and permissible dismantling.

General practical safety: Wear approved PPE; use wheel chocks and stands; keep ignition keys under control; avoid hot, live, pressurised or rotating systems; control fuel vapour and spills; and never bypass guards or interlocks.

Experiment 1

Structural Identification of Chassis and Body

CO1 · Aim: Identify the load-carrying structure and distinguish chassis, frame, subframe and body.

Apparatus

Instructor-approved vehicle or cut-section, wheel chocks, inspection lamp, mirror, PPE and available layout drawing.

Theory

Body-on-frame uses a separate frame; monocoque integrates the body shell as the principal structure.

Safety

Park level, apply brake, chock wheels, isolate key and never enter beneath an unsupported vehicle.

Setup diagram

Mark front/rear, longitudinal members, cross-members, pillars, floor, crumple zones and mounting points on a sketch.

Procedure

1. Record vehicle identity and construction information available.
2. Walk around before underbody inspection.
3. Trace main load paths and aggregate mountings.
4. Classify body-on-frame, monocoque or mixed construction.
5. Sketch and label observed members.

Observation				
Member	**Location**	**Joined by**	**Function**	**Condition**
________	________	Weld/bolt/other	________	________
________	________	________	________	________

Result: The vehicle construction was identified as ________. **Inference:** Principal loads pass through ________.

Precautions / errors: Do not confuse decorative panels with structural members; dirt and covers can hide joints.

Troubleshooting: When structure is inaccessible, use approved diagrams and visible mount relationships; do not remove safety-critical panels without permission.

Viva: Why is a subframe used?

It supports selected aggregates and helps assembly, load distribution, isolation and repair while attaching to the main body or frame.

Record: Vehicle ID ________ · construction ________ · labelled sketch attached □ · instructor signature

Experiment 2

Vehicle Layout Analysis — LMV

CO1 · Aim: Identify aggregate positions and trace power flow in a Light Motor Vehicle.

Apparatus

LMV, chocks, inspection lamp, mirror and manufacturer layout information where available.

Theory

Layout is classified from engine location and driven axle: FWD, RWD, AWD/4WD or another observed arrangement.

Safety

Engine off and cool; do not touch belts, fans, exhaust or high-voltage components.

Setup diagram

Use a top-view outline and place engine, gearbox/transaxle, propeller shaft, differential, fuel tank and driven wheels.

Procedure

1. Identify front, rear and axle lines.
2. Locate engine and transmission.
3. Trace torque path to driven wheels.
4. Identify steering, suspension, brake and tank locations.
5. Draw a block diagram and classify layout.

LMV layout observation

Aggregate	Location	Connection	Observed function
Engine	________	________	Energy conversion
Transmission	________	________	Speed/torque ratio
Final drive	________	________	Driven axle

Result: LMV layout = ________. **Inference:** Power flows ________.

Precautions / errors: Do not infer driven wheels from engine position alone; confirm shafts or transaxle outputs.

Troubleshooting: Covers may hide components; use wheel-side shafts and differential position to verify.

Viva: How do you confirm FWD?

The transmission or transaxle sends torque through drive shafts to the front wheels and no longitudinal propeller shaft is needed for a two-wheel-drive FWD layout.

Record: LMV model ________ · layout ________ · power-flow sketch attached ☐ · signature

Experiment 3

Vehicle Layout Analysis — HMV

CO1 · Aim: Identify heavy-vehicle aggregates and describe the drivetrain configuration.

Apparatus

Stationary HMV or training chassis, chocks, lamp, mirror, PPE and approved chassis drawing.

Theory

HMVs commonly use a separate frame, longitudinal powertrain, propeller shafts and one or more driven rear axles; actual configuration must be observed.

Safety

Use large-vehicle exclusion zone, chock correctly and never inspect air-brake or suspension systems while pressurised unless supervised.

Setup diagram

Draw ladder frame, cab, engine, gearbox, propeller shaft, axles, tanks and cargo/body location.

Procedure

1. Record axle count and wheel arrangement.
2. Identify frame members and cab/body mounting.
3. Trace engine-to-axle torque path.
4. Locate air tanks, fuel tank and major suspension elements without operating them.
5. Classify driven axle arrangement.

HMV configuration

Item	Observed detail	Function
Axle configuration	________	Load and traction
Propeller-shaft sections	________	Torque transmission
Frame type	________	Structural support

Result: HMV configuration = ________. **Inference:** Heavy-duty layout supports ________.

Precautions / errors: Do not stand between vehicle and obstacle; do not mistake a non-driven tag axle for a driven axle.

Troubleshooting: Look for differential housing and propeller-shaft input to identify driven axles.

Viva: Why may an HMV use multiple axles?

To distribute gross load, meet axle-load limits, improve traction or support vehicle stability and braking duties.

Record: HMV type ________ · axle layout ________ · sketch attached ☐ · signature

Experiment 4

Measurement of Vehicle Geometric Dimensions — LMV

CO1 · Aim: Measure wheelbase, track, overhang and ground clearance of an LMV.

Apparatus

LMV, level floor, tape/steel rule, plumb bob or approved laser, chalk, tyre-pressure gauge and chocks.

Theory

Dimensions are meaningful only with defined reference points and stated loading and tyre conditions.

Safety

Vehicle immobilised; no measurement near moving traffic; avoid crawling beneath vehicle.

Setup diagram

Project axle centres and body extremities to floor with plumb lines.

Procedure

1. Check floor, tyre pressure and loading condition.
2. Mark front and rear axle centre projections.
3. Measure wheelbase at both sides and compare.
4. Measure front/rear track using wheel-centre-plane method or approved equivalent.
5. Measure overhangs and minimum specified ground clearance.

LMV dimensions

Dimension	Trial 1 mm	Trial 2 mm	Reported mm	Reference
Wheelbase				Axle centres
Front track				Wheel centre planes
Rear track				Wheel centre planes
Ground clearance				Specified low point

Calculation: Overall-length check = front overhang + wheelbase + rear overhang.

Result: Dimensions recorded above. **Inference:** Repeatability difference = ________ mm.

Precautions / errors: Parallax, tape sag, wrong tyre edge, steering angle and uneven floor are major errors.

Troubleshooting: If left/right wheelbase differs unexpectedly, repeat projections before concluding structural misalignment.

Viva: Why record tyre pressure?

Tyre deflection changes ride height and can affect ground-clearance and reference measurements.

Record: Loading condition ________ · tyre pressure ________ · signature

Experiment 5

Measurement of Vehicle Geometric Dimensions — HMV

CO1 · Aim: Interpret and measure geometric dimensions of a Heavy Motor Vehicle.

Apparatus

HMV/training chassis, long tape, plumb bob/laser, chalk, level floor, chocks and assistants under instructor control.

Theory

Multi-axle vehicles require the exact axle and wheel references to be identified before measurement.

Safety

Establish exclusion zone; no one moves the vehicle during measurement; use no unsupported underbody access.

Setup diagram

Draw all axle centre lines and identify steering, drive, tag or tandem axles.

Procedure

1. Record axle configuration and loading state.
2. Mark centre line of every axle.
3. Measure specified wheelbase or tandem spacing.
4. Measure track for selected axle using approved reference.
5. Measure overhang and ground clearance without entering unsafe zones.

HMV dimensions			
Reference pair	**Measured mm**	**Repeat mm**	**Difference mm**
Front axle to drive axle			
Tandem spacing			
Selected axle track			
Ground clearance			

Result: HMV dimensions recorded. **Inference:** The stated wheelbase reference was ________.

Precautions / errors: The official PDF incorrectly says “Light Motor Vehicle (HMV)”; this activity is treated as HMV. Never assume which rear axle defines wheelbase.

Troubleshooting: Consult chassis drawing for manufacturer-specific wheelbase definition.

Viva: What is tandem spacing?

The longitudinal centre-to-centre distance between adjacent axles in a tandem group.

Record: Axle configuration ________ · reference definition ________ · signature

Experiment 6

Inspection and Selection of Engine Mounts

CO1 · Aim: Identify mount types, inspect condition and justify a suitable selection concept.

Apparatus

Rigid, rubber and hydraulic mount samples or stationary vehicle, lamp, mirror and manufacturer information.

Theory

Mounts balance load capacity, vibration isolation, damping, movement control, alignment and durability.

Safety

Never remove or loosen a mount without approved engine support and instructor procedure.

Setup diagram

Sketch mount orientation, load direction and torque-reaction direction.

Procedure

1. Identify mount location and construction.
2. Inspect rubber, metal brackets and fasteners visually.
3. Look for sag, separation, cracks, oil attack or hydraulic leakage.
4. Compare type with vibration and load requirement.
5. Record selection criteria without dismantling.

Mount inspection				
Location	Type	Condition	Likely load	Action
________	Rigid/flexible/hydraulic	________	________	Serviceable/review

Result: Mount type ________; condition ________. **Inference:** Selection prioritises ________.

Precautions / errors: Surface dirt is not a crack; compare unloaded and installed geometry only using approved references.

Troubleshooting: Excess vibration may also come from engine misfire, exhaust contact or driveline fault; do not condemn a mount from vibration alone.

Viva: What indicates hydraulic-mount failure?

Fluid leakage, collapse, separation or excessive controlled movement, confirmed against manufacturer inspection criteria.

Record: Mount location/type ________ · defect sketch attached □ · signature

Experimen 7

Identification of Thermodynamic Systems in an Automobile

CO2 · Aim: Classify automotive components as open, closed or approximately isolated for a stated boundary.

Apparatus

Stationary vehicle or component models, system-boundary worksheets and marker.

Theory

Classification depends on whether mass and energy cross the selected boundary during the selected period.

Safety

Engine off and cool unless a supervised demonstration explicitly requires operation.

Setup diagram

Draw a dashed boundary around radiator, tank, battery, cylinder or complete engine and add mass/energy arrows.

Procedure

1. Select component and time period.
2. Draw boundary.
3. List mass crossing.
4. List heat/work/electrical energy crossing.
5. Classify and justify.

System classification

Component/boundary	Mass crossing	Energy crossing	Class	Reason
Running radiator	Coolant/air	Heat/flow work	Open	Mass and energy cross
Fuel tank during sealed rest	Approx. none	Heat possible	Closed approximation	State boundary/time
________	________	________	________	________

Result: Components were classified with stated boundaries. **Inference:** The same component may change classification when operating condition changes.

Precautions / errors: Avoid classifying by component name alone; include vapour vents and electrical work.

Troubleshooting: If uncertain, redraw the boundary and ask whether any molecule crosses it.

Viva: Is a fuel tank always a closed system?

No. During filling, venting or fuel delivery mass crosses its boundary; classification depends on the chosen boundary and time.

Record: Five classified systems and boundary sketches attached □ · signature

Experiment 8

Demonstration of the Zeroth Law and Thermal Equilibrium

CO2 · Aim: Observe temperature stabilisation between a source/body and a measuring instrument.

Apparatus

Instructor-approved warm-water or coolant simulator, two calibrated temperature sensors/thermometers, insulated container and timer.

Theory

A thermometer measures temperature after reaching sufficient thermal equilibrium with the body.

Safety

Use only instructor-approved safe temperatures; no hot pressurised vehicle cooling system.

Setup diagram

Show body A, sensor B and reference sensor C in thermal contact.

Procedure

1. Record initial temperatures.
2. Place sensors as instructed without touching vessel wall.
3. Record at equal time intervals.
4. Continue until readings change negligibly.
5. Compare final readings and discuss lag.

Temperature stabilisation

Time s	Body/reference °C	Sensor °C	Difference K or °C
0			
___			
Final			

Result: Final difference = ________. **Inference:** Measurement requires response time and thermal contact.

Precautions / errors: Sensor immersion depth, wall contact, stirring and calibration affect reading.

Troubleshooting: If readings do not converge, inspect sensor placement, heat loss and calibration status.

Viva: Why wait before reading a thermometer?

The sensor needs time to exchange heat and approach thermal equilibrium with the measured body.

Record: Graph temperature versus time attached □ · signature

Experiment 9

Study of Intensive and Extensive Properties of Engine Oil

CO2 · Aim: Distinguish intensive and extensive properties through engine-oil observations.

Apparatus

Approved clean engine-oil sample, two graduated containers, balance, thermometer and instructor-approved viscosity demonstration.

Theory

Temperature, density and viscosity are intensive; mass and volume are extensive.

Safety

Use clean labelled oil, gloves and spill tray; no skin contact or disposal to drains.

Setup diagram

Divide one sample into two quantities while maintaining approximately equal temperature.

Procedure

1. Record total mass, volume and temperature.
2. Divide into two unequal quantities.
3. Record each mass and volume.
4. Compare temperature and approved flow-time/viscosity indication.
5. Classify each property.

Oil-property observations

Property	Total	Sample A	Sample B	Intensive/extensive
Mass g				Extensive
Volume mL				Extensive
Temperature °C				Intensive
Flow indicator				Intensive at stated T

Result: Properties classified. **Inference:** Viscosity comparison is valid only at a stated temperature and method.

Precautions / errors: Contamination, air bubbles, temperature change and unclean glassware distort results.

Troubleshooting: If density estimates differ widely, repeat mass/volume readings and remove trapped air.

Viva: Why is volume extensive?

Combining otherwise similar oil quantities adds their volumes, so volume depends on system size.

Record: Property classification table completed ☐ · waste returned to approved container ☐ · signature

Experiment 10

Experimental Verification of the First Law of Thermodynamics

CO2 · Aim: Demonstrate conservation of energy using an approved calorimetric/electrical-mechanical model.

Apparatus

Institution-approved insulated water/electrical heater or mechanical-equivalent apparatus, meters, timer and thermometer.

Theory

Energy supplied becomes stored energy plus work and losses. For the defined closed-system model, $Q - W = \Delta U$.

Safety

Use protected low-voltage apparatus and instructor-set limits; avoid exposed live terminals and hot surfaces.

Setup diagram

Show electrical or mechanical input, system boundary, stored thermal energy and loss paths.

Procedure

1. Record mass and initial temperature.
2. Record input voltage/current/time or approved mechanical input.
3. Apply input for specified period.
4. Record final temperature.
5. Calculate input and estimated stored energy using instructor data.
6. Account qualitatively for difference as losses and uncertainty.

Energy balance

Input data	Initial state	Final state	Input energy J	Stored energy J	Difference J
______	______	______			

Result: Input ≈ stored energy + losses. **Inference:** Energy is conserved within measurement uncertainty.

Precautions / errors: Heat loss, sensor lag, meter uncertainty and unaccounted apparatus heat capacity cause mismatch.

Troubleshooting: Check units, time base and electrical power calculation before blaming apparatus.

Viva: Why are measured input and stored energy rarely identical?

Some energy heats the container and surroundings, and measurements have uncertainty; all terms must be included for a complete balance.

Record: Formula and substitutions shown ☐ · percentage difference ________ · signature

Experiment 11

Experimental Verification of the Second Law of Thermodynamics

CO2 · Aim: Demonstrate heat flow direction and the necessity of a heat sink in an engine model.

Apparatus

Approved hot/cold reservoirs or heat-engine demonstration model, temperature sensors and timer.

Theory

Spontaneous heat transfer is from higher to lower temperature, and a cyclic heat engine rejects heat.

Safety

Use safe instructor-approved temperatures and insulated handling tools.

Setup diagram

Hot source QH → engine/model → work output and QL → cold sink.

Procedure

1. Record source and sink temperatures.
2. Establish approved contact or run demonstration.
3. Observe temperature changes and any work output.
4. Identify rejected heat path.
5. Explain why η cannot be 100%.

Second-Law observation

Time	Hot-side T	Cold-side T	Direction observed	Heat-sink evidence
Initial				
Final				

Result: Heat flowed ________. **Inference:** A radiator/exhaust/surroundings acts as a sink; 100% cyclic conversion is impossible.

Precautions / errors: Do not use “heat disappears”; identify where it goes.

Troubleshooting: Small temperature difference gives weak observation; use only instructor-approved conditions.

Viva: Does the First Law alone predict heat-flow direction?

No. The First Law conserves energy; the Second Law supplies direction and feasibility limits.

Record: Energy-flow diagram attached □ · signature

Experimen 12

Experimental Verification of the Third Law of Thermodynamics

CO2 · Aim: Discuss absolute zero and use cooling data to show approach to a limit rather than practical attainment.

Apparatus

Instructor-provided cooling curve, simulation or safe cooling demonstration; thermometer and graph sheet.

Theory

As temperature approaches 0 K, entropy of a perfect crystal approaches a minimum; absolute zero is unattainable by a finite practical sequence.

Safety

No cryogenic material unless the institution has dedicated facilities and trained supervision. A dataset/simulation is preferred.

Setup diagram

Plot temperature against time or cooling stages, with 0 K shown as an asymptote.

Procedure

1. Review kelvin scale and convert supplied temperatures.
2. Plot cooling data.

3. Observe decreasing rate or increasing effort per temperature reduction.
4. Extrapolate conceptually toward 0 K without claiming attainment.
5. Relate to real automotive thermal processes.

Cooling dataset			
Stage/time	**Temperature °C**	**Temperature K**	**Reduction from prior stage K**
1			—
2			
3			

Result: The curve approaches a lower limit. **Inference:** The Third Law is a theoretical limiting statement, not a classroom claim of reaching 0 K.

Precautions / errors: −273.15 °C corresponds to 0 K; kelvin is not written with a degree symbol.

Troubleshooting: If the graph appears linear, do not over-interpret a short dataset; discuss the conceptual limitation.

Viva: Can an automobile cooling system reach absolute zero?

No. It operates near ambient and engine temperatures, and absolute zero cannot be reached by a finite practical process.

Record: Cooling graph attached ☐ · conclusion mentions asymptotic limit ☐ · signature

Experiment 13 — Study of Different Tools Used in Automobile Workshop

CO3 · Aim: Identify common hand, power and precision tools and select them correctly.

Apparatus

Approved display of spanners, sockets, screwdrivers, pliers, torque wrench, puller, jack/stand samples, vernier caliper, micrometer, bore gauge and PPE.

Theory

Correct tool geometry, range and condition prevent component damage and injury.

Safety

Power tools disconnected for identification; no damaged insulated handles, mushroomed punches or makeshift extensions.

Setup diagram

Group tools by fastening, holding, striking, pulling, lifting and measurement.

Procedure

1. Read tool name, size/range and condition.
2. Identify correct application.
3. Demonstrate safe grip and storage without live operation.
4. Check precision instrument zero.
5. Record misuse risk.

Tool identification				
Tool	**Range/size**	**Application**	**Pre-use check**	**Wrong-use hazard**
Torque wrench	________	Specified tightening	Calibration/setting	Over/under-tightening
________				

Result: Tools identified and classified. **Inference:** Selection depends on fastener, access, torque and precision.

Precautions / errors: An adjustable spanner is not a universal substitute; a torque wrench is not a breaker bar.

Troubleshooting: If a fastener rounds, stop and reassess size, engagement and tool condition.

Viva: Why check instrument zero?

A zero offset produces systematic error in every measurement unless corrected.

Record: Ten tools sketched/identified ☐ · signature

Experiment 14

Identification of Major Engine Components

CO3 · Aim: Classify internal and external engine parts and explain each function.

Apparatus

Sectioned engine or dismantled training parts, trays, labels, PPE and approved component drawing.

Theory

Components form structural, gas-exchange, reciprocating, rotating, lubrication, cooling, ignition or injection systems.

Safety

Use lifting aids for heavy parts; protect sharp edges; do not rotate assembly unless authorised and hands are clear.

Setup diagram

Arrange parts in functional sequence from cylinder head to crankshaft.

Procedure

1. Identify block and head.
2. Locate piston, rings, pin, rod and crankshaft.
3. Identify valves and timing components visible.
4. Trace oil/coolant passages where sectioned.
5. Label and state function and motion.

Engine component record

Part	Internal/external	Function	Motion	Observed feature
Piston	Internal	Receives gas force	Reciprocating	Ring grooves
______				

Result: Major components identified. **Inference:** Force path is piston → rod → crankshaft.

Precautions / errors: Keep matched caps and parts identified; never mix bearing shells or orientation marks.

Troubleshooting: Use oil holes, bearing surfaces and joint features to distinguish visually similar parts.

Viva: What is the cylinder head’s function?

It closes the cylinder, forms the chamber and carries gas passages, valves and spark plug or injector as applicable.

Record: Labelled engine-section sketch attached □ · signature

Experiment 15

Measurement of Engine Geometrical Parameters

CO3 · Aim: Measure bore and stroke references and calculate swept volume and compression ratio from approved data.

Apparatus

Training cylinder/engine, bore gauge or internal micrometer, vernier/depth gauge, outside micrometer, calculator and clean cloth.

Theory

$V_s = \pi D^2 L/4$ and $r = (V_s+V_c)/V_c$.

Safety

Assembly secured; no forced rotation; protect precision instruments from impact and dirt.

Setup diagram

Show bore measured perpendicular to axis and stroke between TDC and BDC references.

Procedure

1. Clean measuring surfaces and check zero.
2. Measure bore at approved heights/directions.
3. Determine stroke using crank/piston reference or supplied drawing.
4. Record cylinder count and clearance-volume data supplied.
5. Calculate and verify units.

Engine geometry				
Parameter	**Reading 1**	**Reading 2**	**Reported**	**Unit**
Bore D				mm
Stroke L				mm
Clearance volume Vc				cm^3

Calculation: Vs = ________; total displacement = ________; r = ________.

Result: Engine geometry calculated. **Inference:** Differences by direction can indicate wear, but diagnosis needs specified limits.

Precautions / errors: Keep instrument square; convert mm^3 to cm^3 correctly: 1000 mm^3 = 1 cm^3.

Troubleshooting: An unstable bore reading usually means tilt, dirt, poor zero setting or inconsistent contact.

Viva: Why measure bore in more than one direction?

To detect ovality or directional wear and improve confidence in the reported diameter.

Record: Full calculation with units attached □ · signature

Experiment 16

Demonstration of Four-Stroke Cycle — Petrol Engine

CO3 · Aim: Demonstrate suction, compression, power and exhaust and the spark-plug role.

Apparatus

Sectioned hand-turned SI engine model, stroke chart and PPE.

Theory

The four-stroke SI cycle uses two crank revolutions; spark initiates combustion near the end of compression.

Safety

Use only hand-turning point provided; ignition/fuel disabled; fingers away from gears, valves and flywheel.

Setup diagram

Mark TDC, BDC, inlet valve, exhaust valve and spark plug.

Procedure

1. Set model at TDC before suction.
2. Turn in correct direction and observe inlet opening/piston descent.
3. Continue compression with both valves closed.
4. Identify spark event near TDC.
5. Observe power descent and exhaust return.
6. Count 720° for one cycle.

Petrol four-stroke observation

Stroke	Piston direction	Inlet	Exhaust	Event
Suction				
Compression				
Power				
Exhaust				

Result: Four strokes and spark event demonstrated. **Inference:** Only the power stroke delivers positive expansion work in the idealised sequence.

Precautions / errors: Real valve timing includes lead, lag and overlap; do not force a basic model to exact dead-centre switching.

Troubleshooting: If valve sequence seems wrong, confirm rotation direction and timing marks.

Viva: How many crank revolutions per cycle?

Two revolutions, equal to 720°.

Record: Stroke table and valve-state diagram completed ☐ · signature

Experiment 17

Demonstration of Four-Stroke Cycle — Diesel Engine

CO3 · Aim: Demonstrate the CI four-stroke sequence and distinguish injection from spark ignition.

Apparatus

Sectioned hand-turned CI engine/model, injector sample and cycle chart.

Theory

Air is compressed to high temperature; fuel injection near compression end leads to self-ignition.

Safety

Fuel system depressurised and disabled; never expose skin to injection spray.

Setup diagram

Label injector, inlet/exhaust valves, piston and compression chamber.

Procedure

1. Observe air-only suction.
2. Turn through high-compression stroke.
3. Identify injector timing point.
4. Follow expansion/power.
5. Follow exhaust stroke.
6. Compare with petrol model.

Diesel cycle observation

Stroke	Charge	Valve state	Injection/combustion event
Suction	Air		None
Compression	Air		Injection near end
Power	Combustion products		Expansion
Exhaust	Products		Expelled

Result: CI cycle demonstrated. **Inference:** Ignition source is compression-heated air, not a spark plug.

Precautions / errors: Glow plugs, where fitted, assist starting and are not normal-cycle spark plugs.

Troubleshooting: If injector cannot be seen, use the fuel line/head position and section drawing.

Viva: What enters during diesel suction?

Air only in the basic diesel cycle.

Record: SI–CI comparison completed ☐ · signature

Experiment 18

Demonstration of Two-Stroke Cycle — Petrol Engine

CO3 · Aim: Illustrate two-stroke operation, port functions and scavenging.

Apparatus

Sectioned hand-turned two-stroke petrol model, port-timing chart and safe flow arrows/markers.

Theory

One cycle occurs per crank revolution; piston position controls inlet, transfer and exhaust ports in the common model.

Safety

Fuel and ignition disabled; keep hands clear and rotate only at approved point.

Setup diagram

Label inlet, transfer and exhaust ports and crankcase volume.

Procedure

1. Rotate piston upward and observe compression above and induction below.
2. Identify ignition near TDC.
3. Rotate downward through expansion.
4. Observe exhaust-port opening.
5. Observe transfer-port opening and scavenging path.
6. Record port sequence and overlap.

Two-stroke port timing observation

Piston region	Inlet	Transfer	Exhaust	Main process
Near TDC/upstroke				Compression/induction
Downstroke				Power/crankcase compression
Near BDC				Exhaust/scavenging

Result: Port sequence and scavenging demonstrated. **Inference:** Port overlap can create fresh-charge short-circuiting.

Precautions / errors: Do not describe the two strokes simply as suction and exhaust; combined processes occur above and below the piston.

Troubleshooting: Use piston crown relative to port edges to establish opening order.

Viva: What is scavenging?

The process of clearing exhaust gas and filling the cylinder with fresh charge.

Record: Port-timing sketch attached ☐ · signature

Experiment 19

Mapping the Fuel Journey — Flowchart

CO4 · Aim: Summarise fuel processing from crude extraction to retail and vehicle tank.

Apparatus

Process cards, official/supplementary refinery information, chart paper or drawing software.

Theory

Fuel passes through extraction, transport, refining, treating, blending, storage, distribution and dispensing with quality control at each stage.

Safety

Desk activity or authorised visit only; no access to restricted process or unloading areas.

Setup diagram

Start → crude extraction → transport → refinery → terminal → tanker → station tank → dispenser → vehicle tank → end.

Procedure

1. Arrange stages in sequence.
2. Add inputs, outputs and quality checks.
3. Mark high-risk transfer points.
4. Use correct process/decision shapes and arrows.
5. Review for missing return, vent and documentation paths.

Fuel-journey stage record

Stage	Main operation	Quality check	Safety/environment control

Stage	Main operation	Quality check	Safety/environment control
Refinery	Separate/treat/blend	Specification testing	Containment and process control
Retail	Store/dispense	Grade/quantity/contamination	Vapour, spill and ignition control

Result: A complete structured flowchart was produced. **Inference:** Product integrity can fail at any transfer interface.

Precautions / errors: Distillation alone does not create finished fuel; include treating, blending and testing.

Troubleshooting: If arrows cross, reorganise stages into production, distribution and retail lanes.

Viva: Why is traceability needed?

It links batches, specifications and transfers so contamination or quality failures can be identified and contained.

Record: Final fuel-journey flowchart attached □ · signature

Experiment 20

Fuel Identification and Visual Inspection

CO4 · Aim: Observe physical characteristics of labelled petrol, diesel and biofuel samples without treating appearance as definitive identification.

Apparatus

Instructor-prepared sealed transparent labelled samples or approved photographs, spill tray, PPE and documentation.

Theory

Colour, clarity, odour and apparent viscosity can vary with dye, additive, temperature and contamination; verified labels and approved tests are authoritative.

Safety

No open containers, smelling directly, ignition source, skin contact or student decanting. Use ventilation and minimal sealed samples.

Setup diagram

Place sealed samples in secondary containment with identity hidden only if instructor can preserve safe traceability.

Procedure

1. Read safety and label information.

2. Observe colour and clarity through container.
3. Compare bubble/flow behaviour only if instructor permits sealed movement.
4. Record possible contamination signs.
5. Reveal/confirm identity from documentation.

Visual inspection				
Sample ID	**Labelled fuel**	**Colour/clarity**	**Relative flow**	**Contamination signs**
A				
B				

Result: Observations recorded; identity confirmed by label/documentation. **Inference:** Visual inspection supports but cannot certify fuel identity.

Precautions / errors: Never use direct smell or flame; water can appear as separate phase or haze but needs approved confirmation.

Troubleshooting: If label is missing, quarantine sample rather than guessing.

Viva: Can petrol and diesel always be identified by colour?

No. Colour can vary with dyes, grade and contamination; documentation and approved testing are required.

Record: Samples remained sealed ☐ · disposal/return confirmed ☐ · signature

Experiment 21

Study of Vehicle Fuel Tank and Filling System

CO4 · Aim: Identify cap, filler neck, tank, venting and fuel-supply elements.

Apparatus

Depressurised training tank or stationary vehicle, lamp, mirror, system diagram and PPE.

Theory

The filling system must accept fuel, prevent leakage/overfill and manage vapour or pressure while supplying clean fuel.

Safety

No opening, draining or energising unless specifically approved; no sparks, hot work or unventilated inspection.

Setup diagram

Cap → filler neck → anti-spill/inlet → tank → pump/pick-up → line; vent → EVAP/canister where fitted.

Procedure

1. Identify fuel type and cap label.
2. Trace accessible filler path.
3. Locate tank and mountings.
4. Identify vent/vapour and supply paths from diagram/visible lines.
5. Inspect for damage or leakage without disturbing connections.

Fuel-system observation				
Component	**Location**	**Function**	**Condition**	**Connection traced**
Cap		Seal/pressure interface		
Vent path		Pressure/vapour management		
Tank		Storage		

Result: Tank and filling system identified. **Inference:** Venting is necessary but must control vapour emissions and liquid escape.

Precautions / errors: Do not confuse supply, return and vapour lines; follow routing and diagram.

Troubleshooting: A filling complaint can involve cap, filler restriction, vent fault or tank valve; diagnosis needs approved tests.

Viva: Why can a sealed tank not be perfectly unvented?

Fuel consumption and temperature changes create pressure or vacuum; a controlled vent/vapour system manages them safely.

Record: Fuel-system block diagram attached □ · signature

Experiment 22 — Study of Safety Equipment at a Petrol Pump

CO4 · Aim: Identify fuel-station safety equipment, controls and safe operating zones.

Apparatus

Authorised site visit or training layout, observation checklist, PPE/high-visibility wear as required.

Theory

Protection combines prevention, detection, isolation, containment, extinguishing, signage and emergency response.

Safety

Follow site induction and staff instructions; no phone use, photography or entry into restricted zones unless expressly permitted.

Setup diagram

Mark emergency stop, extinguishers, sand/spill kit, vents, unloading point, isolation and evacuation route.

Procedure

1. Receive authorised safety briefing.
2. Observe dispenser and emergency controls from safe area.
3. Locate extinguishers and note class/inspection status without operating.
4. Identify spill-control and warning signage.
5. Trace evacuation and emergency communication.

Safety-equipment checklist

Equipment/control	Location	Purpose	Inspection/status observed
Emergency stop		Stop dispensing/power as designed	
Extinguisher		Incipient fire response by trained person	
Spill kit		Contain spill	

Result: Safety controls identified. **Inference:** Equipment is effective only with access, maintenance, training and a clear emergency plan.

Precautions / errors: Never operate emergency equipment as a demonstration without site authority.

Troubleshooting: If an item is inaccessible or inspection status unclear, report to responsible staff; do not relocate it.

Viva: What should a student do on discovering a fuel spill?

Raise the alarm, avoid ignition and contact, keep away, and follow site staff/emergency instructions rather than attempting an untrained cleanup.

Record: Site/virtual layout ________ · induction completed ☐ · signature

Experimen 23

Flash Point and Fire Point Demonstration — Safety Test

CO4 · Aim: Observe the distinction between flash and sustained fire points using approved apparatus or recorded demonstration.

Apparatus

Named institution-approved standard apparatus, specified safe training liquid/sample, calibrated thermometer and fire-control equipment—or a recorded certified demonstration.

Theory

Flash point is momentary vapour ignition; fire point is sustained burning under specified method conditions.

Safety

Instructor performs the test behind suitable controls. Students do not handle flame or fuel. Exact quantity, heating rate and test interval come only from the approved method/manual.

Setup diagram

Show cup, temperature sensor, controlled heater, test-flame/application system, shield and extinguisher.

Procedure

1. Identify apparatus and method.
2. Confirm sample and emergency controls.
3. Observe controlled heating.
4. Record instructor-announced flash temperature.
5. Where the approved method includes it, record sustained-fire temperature.
6. Allow safe shutdown and cooling.

Safety-test observation

Sample/method	Flash point °C	Fire point °C	Observation	Instructor initials

Result: Flash point ________; fire point ________. **Inference:** Fire point is normally above flash point for the same method/sample.

Precautions / errors: Apparatus, cup type, heating rate, contamination and atmospheric conditions affect results. Never improvise.

Troubleshooting: An anomalous result requires test invalidation and method review—not repeated uncontrolled flame application.

Viva: Is flash point the same as auto-ignition temperature?

No. Flash point uses an external ignition source under a test method; auto-ignition occurs without an external spark or flame.

Record: Approved method named ________ · observation only ☐ · signature

Experiment 24

Understanding BS-VI and Fuel Labelling

CO4 · Aim: Interpret BS-VI fuel labels, grade information and biofuel-blend markings.

Apparatus

Photographs or clean examples of fuel labels/caps, sample invoices or specification sheets and vehicle owner information.

Theory

Fuel family, octane/cetane context, blend level and emission-stage compatibility are separate pieces of information.

Safety

Use documents or clean inactive examples; no sampling at a live dispenser.

Setup diagram

Label fields: petrol/diesel, grade, RON where shown, ethanol/biodiesel blend, BS-VI/compliance information and warnings.

Procedure

1. Read vehicle fuel requirement.
2. Identify fuel family.
3. Record grade and blend marking.
4. Compare with vehicle approval.
5. Identify misfuelling and contamination risks.

Label interpretation

Label/sample	Fuel family	Grade	Blend	Compatible?	Reason
A					
B					

Result: Labels interpreted against vehicle requirements. **Inference:** BS-VI does not remove the need to match fuel type, grade and blend.

Precautions / errors: Do not assume a higher octane grade gives more energy; do not put petrol in diesel or vice versa.

Troubleshooting: If compatibility is uncertain, do not fill; consult manufacturer documentation.

Viva: What does E20 indicate?

Petrol containing approximately 20% ethanol by volume, subject to the applicable specification; the vehicle must be approved for it.

Record: Three labels analysed □ · compatibility source noted □ · signature

Malayalam support: Practical record-□ observed data □□□□□□ □□□□□□. Instructor □□□□□ exact procedure, service limit, test method □□□□□□ □□□□□□□□□. Guess □□□□□ value record □□□□□□□□□□□ engineering evidence □□□□.

ASSESSMENT PRACTICE

Expected / Practice Questions with Answers

These are handbook practice questions, not official SITTTR previous questions. Answers show the minimum technically correct points; diagrams, units and safety conditions must be added where required.

Module I — Automobile architecture

Objective self-check

1. The distance between axle centre lines is ________. **Answer:** wheelbase.
2. Vehicle weight ready for operation without payload is ________. **Answer:** kerb weight under the applicable definition.
3. A front-engine vehicle with propeller shaft to rear differential is ________. **Answer:** front-engine RWD.
4. A fluid-filled vibration-control mount is a ________. **Answer:** hydraulic mount.

Model-answer points

Explain monocoque: integrated body shell; panels and reinforcements carry loads; no complete separate ladder frame; subframes may still be used.

Why lower CG helps: reduces lateral/longitudinal load transfer and rollover tendency for comparable track and dynamics.

Descriptive **Explain the evolution of automobiles and its engineering significance.**

Cover steam/early electric/IC development, mass production, steel bodies, braking and tyre improvements, electronics, emission control, hybrid/electric systems and the resulting changes to design, testing, service and safety.

Application **A vehicle vibrates at idle after a mount replacement. Give a logical inspection sequence.**

Confirm correct mount part/orientation and torque; inspect preload/alignment and all mounts; check exhaust/body contact; check engine misfire and idle condition; compare movement to manufacturer criteria. Do not assume the new mount alone is defective.

Module II — Thermodynamics

Objective self-check

1. Mass and energy cross an ________ system. **Answer:** open.
2. Pressure is generally an ________ property. **Answer:** intensive.
3. $Q - W = \Delta U$ expresses the ________ Law. **Answer:** First.
4. 0 K equals ________ °C. **Answer:** approximately −273.15 °C.

Model-answer points

Zeroth Law: thermal equilibrium is transitive; establishes temperature measurement.

Second Law: direction, entropy and heat rejection; no 100% cyclic heat engine.

Numerical **A closed system receives 2.2 kJ heat and does 0.9 kJ work. Find ΔU.**

$\Delta U = Q - W = 2.2 - 0.9 =$ **+1.3 kJ**.

Numerical **A heat engine receives 5 MJ and rejects 3.4 MJ. Find work and efficiency.**

$W = 5 - 3.4 = 1.6$ MJ. $\eta = 1.6/5 = 0.32 =$ **32%**.

Higher Order **Can a radiator be called an open and a closed system?**

Yes, depending on boundary. A control volume around the radiator has coolant and air mass flow and is open. A fixed trapped coolant mass considered over a selected interval may be treated as closed while exchanging heat.

Module III — Engine basics

Objective self-check

1. Piston position nearest the head is ________. **Answer:** TDC.
2. A four-stroke cycle needs ________ crank revolutions. **Answer:** two.
3. Petrol-engine ignition is generally ________. **Answer:** spark ignition.
4. Clearing exhaust with fresh charge is ________. **Answer:** scavenging.

Model-answer points

Compression ratio: total cylinder volume at BDC divided by clearance volume at TDC; dimensionless.

CI operation: air induction, high compression, injection, ignition delay/self-ignition, expansion and exhaust.

Numerical **Find total displacement for 3 cylinders, bore 70 mm and stroke 75 mm.**

Per cylinder = $\pi/4 \times 70^2 \times 75$ = 288,633 mm³ = 288.6 cm³. Total = 3×288.6 = **865.9 cm³**.

Comparison **Compare two-stroke and four-stroke petrol engines.**

Two-stroke: cycle in one revolution, ports and scavenging, theoretically one power event/revolution, simpler but lubrication/emission and charge-loss challenges. Four-stroke: two revolutions, valve-operated separate strokes, better gas-exchange and lubrication control in common designs.

Troubleshooting **An engine has low compression. Name possible mechanical causes without giving a final diagnosis.**

Valve leakage/timing error, worn or broken rings, cylinder wear, head-gasket leakage, piston damage or incorrect test procedure. Confirm using approved compression/leak-down and timing checks.

Module IV — Fuels

Objective self-check

1. Petrol knock resistance is indicated by ________. **Answer:** octane number.
2. Diesel ignition quality is indicated by ________. **Answer:** cetane number.
3. Sustained burning temperature under a method is ________. **Answer:** fire point.

4. E20 denotes approximately ________ ethanol. **Answer:** 20% by volume.

Model-answer points

Fuel station safety: ignition control, emergency shutdown, spill containment, ventilation/vapour control, unloading discipline, trained response and maintained extinguishers.

BS-VI context: low-sulphur compatible fuels plus vehicle control and after-treatment.

Numerical **Calculate chemical energy in 1.2 kg fuel of CV 42.5 MJ/kg.**

$E = m \times CV = 1.2 \times 42.5 =$ **51 MJ**.

Application **A fuel sample is clear and yellowish. Can it be declared diesel?**

No. Appearance is insufficient. Use traceable label/documentation and an approved test; quarantine an unknown sample.

Safety **What is wrong with a student performing a flash-point test in an open container?**

It ignores the specified apparatus/method, heating control, vapour containment, ignition application and emergency safeguards, producing unsafe and invalid results.

EXAMINATION REHEARSAL

Practice Model Paper — not an official SITTTR model question paper

The supplied official syllabus does not specify a model-paper pattern. This practice set is organised by question type, not claimed marks.

Part A — Very Short Answer

1. Define wheelbase.
2. State the Zeroth Law.
3. Define TDC.
4. What does BS-VI mean in this course?
5. Differentiate flash and fire points in one sentence.
6. Define compression ratio.
7. Name two intensive properties.
8. What is a hydraulic engine mount?

Part B — Short Answer

1. Compare FWD and RWD.
2. Classify radiator, trapped cylinder gas and ideal insulated container.
3. Explain the four-stroke petrol sequence.
4. Distinguish octane and cetane numbers.
5. Explain tank venting.
6. List workshop tool-selection rules.

Part C — Descriptive / Diagram

1. Draw and explain a front-engine RWD layout.
2. Explain all four thermodynamic laws with automotive examples.
3. Draw SI/CI comparison and engine component section.
4. Draw the crude-to-vehicle fuel journey and filling-station safety layout.

Part D — Numerical / Procedure

1. Calculate displacement from given bore, stroke and cylinder count.
2. Calculate ΔU from heat and work.
3. Calculate heat-engine efficiency.
4. Write a complete geometric-dimension measurement procedure.
5. Write a safe flash/fire-point observation procedure.

Model-answer checklist

- Use exact definitions and units.
- Show formula, substitution and final answer.
- Label all diagram arrows and components.
- State boundary/sign convention for thermodynamics.
- State reference and loading condition for dimensions.
- Include isolation, PPE, approved apparatus and instructor authority in practical answers.

LAST-MINUTE CONSOLIDATION

Quick Revision

Module I

Structure → layout → dimensions → weight/CG → mounts.

Trap: engine front ≠ automatically FWD.

Module II

Boundary → properties → laws → processes → ideal cycles.

Trap: adiabatic does not mean constant temperature.

Module III

Parts → geometry → four-stroke SI → four-stroke CI → two-stroke.

Trap: diesel suction is air only in the basic cycle.

Module IV

Extract → refine → blend → store → dispense → label/safety.

Trap: octane is not calorific value.

One-look comparison

Pair	First	Second
Kerb / GVW	Ready-to-operate without payload under definition	Gross specified vehicle condition
Open / closed	Mass and energy may cross	No mass; energy may cross
SI / CI	Spark ignition	Compression ignition after injection
Octane / cetane	Petrol knock resistance	Diesel ignition quality
Flash / fire	Momentary vapour ignition	Sustained burning under method

Common mistakes

- Mixing tyre outside width with track.
- Calling an ideal cycle an exact real-engine process.
- Using °C in Carnot temperature ratios.
- Forgetting the π/4 factor when bore is diameter.
- Calling glow plug a diesel spark plug.
- Identifying unknown fuel by smell or colour.
- Writing practical values that were not observed.

Exam and practical checklist

☐ Definitions exact
☐ Units present

☐ Diagram labelled

☐ Boundary/reference stated
☐ Assumptions stated
☐ Answer checked

☐ PPE/isolation
☐ Approved apparatus
☐ Result and inference separate

TECHNICAL VOCABULARY

Glossary

Key terms

Term	Concise meaning
Automobile	Self-propelled road vehicle for people or goods.
Chassis	Load-carrying system locating major vehicle aggregates.
Monocoque	Body shell acting as the main structure.
Wheelbase	Longitudinal distance between axle centre lines.
Track	Distance between wheel centre planes on one axle.
Overhang	Axle centre line to corresponding vehicle extremity.
Kerb weight	Ready-to-operate vehicle weight/mass under the applicable definition without payload.
GVW	Gross vehicle weight/mass at the specified loaded condition.
Centre of gravity	Point through which resultant vehicle weight is considered to act.
System	Matter or region selected for thermodynamic study.
Boundary	Surface separating system and surroundings.
Enthalpy	Property $H = U + pV$.
Entropy	Thermodynamic state property associated with energy dispersal and irreversibility.
TDC / BDC	Nearest/farthest piston dead-centre positions relative to head.
Bore	Cylinder diameter.
Stroke	Piston travel between dead centres.
Swept volume	Volume displaced by piston during one stroke.
Clearance volume	Volume above piston at TDC.
Compression ratio	Maximum cylinder volume divided by minimum volume.

Term	Concise meaning
Scavenging	Removal of exhaust and replacement with fresh charge.
Octane number	Petrol knock-resistance indicator.
Cetane number	Diesel ignition-quality indicator.
Volatility	Tendency of a liquid to vaporise.
Viscosity	Resistance of a fluid to flow.
BS-VI	Bharat Stage VI emission/fuel-quality context.

SOURCE DISCIPLINE

References

Official curriculum source: Supplied SITTTR Kerala Diploma Curriculum Revision 2026 syllabus PDF, Course 1051 Basic Automobile Engineering.

Official textbooks, reference books, websites and digital resources: Not specified in the supplied official syllabus PDF.

Supplementary references used for explanatory context

- Current official manufacturer corporate and plant information for Hero MotoCorp, Bajaj Auto, Maruti Suzuki, Tata Motors, Ashok Leyland and Mahindra & Mahindra; verify current allocations because they can change.
- Government/industry official BS-VI fuel-quality information and IndianOil fuel specification material for low-sulphur context.
- Institution-approved vehicle service manuals, measuring-instrument instructions and laboratory test methods for actual practical work.

Source rule: A supplementary reference may explain an official syllabus topic, but it does not alter official course outcomes, hours, activities or assessment data.

HANDBOOK COMPLETE

Course 1051 Completion Check

All four official modules and all 24 listed practical activities are represented. Use the print controls to reveal every answer and practical record section in one A4 handbook.